PROCESS CELL: PC_ANT_PACK — Packaging Process Cell (ANT — Antwerp | EU Site)

SITE HIERARCHY — PC_ANT_PACK

PC_ANT_PACK → UNIT_ANT_PACK → EM_ANT_PACK

FLOW WRAPPER ZONE

PALLETIZER ZONE

TT 001 — WRAP_SEAL_TEMP_DEGC
PT 002 — FLOW_WRAPPER_PRESSURE_
ST 003 — WRAP_SPEED_MPM
IT 004 — WRAP_MOTOR_CURRENT_A

ST 011 — PALL_SPEED_RPM
CT 012 — PALL_COUNT_PALLETS
WI 013 — PALL_WEIGHT_KG
IT 014 — PALL_MOTOR_CURRENT_A

FILM REEL — LDPE FILM

EQ_ANT_018

ANT Flow Wrapper

(Flow Wrapper)

SEAL JAWS

Mfr: Festo

OS 018 — WRAP_OPERATING_STATE
AS 019 — WRAP_ALARM_STATE

FROM PC_ANT_COOL

(COOLED COOKIES)

OUTFEED / ACCUMULATION CONVEYOR

EQ_ANT_019

ANT Palletizer

(Palletizer)

PALLET

Mfr: Bosch

OS 018 — PALL_OPERATING_STATE
AS 019 — PALL_ALARM_STATE

TO WAREHOUSE / SHIPPING

(PACKAGED PALLETS)

REJECT/WASTE

WT 005 — WRAP_WEB_TENSION_N
CT 006 — WRAP_COUNT_RATE_PCS_MI
FT 007 — WRAP_FILM_FEED_M_MIN

LT 015 — PALL_LAYER_COUNT
RT 016 — PALL_REJECT_RATE_PCS_M

PLC CABINET — UNIT_ANT_PACK / EM_ANT_PACK

CM_COOKIE_036 — Flow Wrapper — Seal 5 — *[BatchCtrl]* — PLC-04/D81235

CM_COOKIE_044 — Flow Wrapper — Seal 6 — *[SEQ]* — PLC-12/D81243

CM_COOKIE_052 — Flow Wrapper — Seal 7 — *[MotorCtrl]* — PLC-04/D81251

VARIABLE FREQ. DRIVES — EM_ANT_PACK

EQ_AUX_012	EQ_AUX_026	EQ_AUX_040	EQ_AUX_054
VFD/PI/z	VFD/E-HI	VFD/PI/z	VFD/Danfoss
Motor Drive 12	Motor Drive 26	Motor Drive 40	Motor Drive 54

VFDs: EQ_AUX_012 · EQ_AUX_026 · EQ_AUX_040 · EQ_AUX_054

NOTES:

1. Process cell PC_ANT_PACK (ANT — Antwerp | EU Site): FROM PC_ANT_COOL → Flow Wrapper → Palletizer → Warehouse.

2. Flow Wrapper EQ_ANT_018 (Festo) wraps cookies in LDPE film; sealing controlled by PLC.

3. Palletizer EQ_ANT_019 (Bosch) builds and wraps pallet loads.

4. VFDs (EQ_AUX_012, EQ_AUX_026, EQ_AUX_040, EQ_AUX_054) provide variable speed for wrappers and conveyors.

5. TS_-_PACK_FLOW_WRAPPER_PRESSURE_BAR is the only explicitly modelled CDF time-series for this cell.

6. State encoding: 0=STOPPED, 1=RUNNING, 2=IDLE, 3=STARTUP, 4=ALARM, 5=MAINTENANCE.

LEGEND

- TT — Seal Bar Temperature
- PT — Seal Jaw Pressure
- ST — Speed Transmitter
- IT — Motor Current Indicator
- WT — Web Tension
- CT — Count Rate
- FT — Film Feed Rate
- WI — Pallet Weight
- LT — Layer Count
- RT — Reject Rate
- OS — Operating State
- AS — Alarm State